

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

10 August 2026 | 08:49 PM

Session ID: 58

Patient: Eric

Modules: X-Ray Analysis

X-Ray Analysis

Detected Region: Chest (Thorax)

No significant findings detected.

■ **Urgency Level: Urgent — Same day attention required**

Recommended Tests

- Sputum smear microscopy (AFB) or GeneXpert MTB/RIF to rule out tuberculosis
- Complete blood count with differential, blood culture, and CRP/ESR to assess infection severity
- Chest X-ray with lateral view for better localization and characterization of infiltrates
- BNP or echocardiography to assess cardiac function if pulmonary edema suspected
- Blood glucose, renal function tests (urea, creatinine) to identify underlying comorbidities

Suggested Treatment

- Empiric broad-spectrum antibiotics (amoxicillin-clavulanate or fluoroquinolone) pending culture results if pneumonia suspected
- Oxygen therapy to maintain SpO2 >90% if hypoxemic
- Supportive care with fluid management tailored to the underlying etiology
- Anti-TB therapy initiation if TB confirmed, following national TB guidelines

Next Steps

- Obtain detailed clinical history including fever duration, cough character, sputum production, and risk factors
- Perform clinical examination including vital signs, oxygen saturation, and respiratory assessment
- Urgently obtain sputum samples for TB screening given prevalence in West Africa
- Assess for signs of respiratory distress or hypoxemia requiring immediate oxygen therapy
- Review cardiac risk factors and obtain ECG if pulmonary edema is being considered

Summary: This chest X-ray shows diffuse bilateral pulmonary infiltrates with alveolar opacities concerning for pneumonia as the primary consideration. Tuberculosis and pulmonary edema remain important differential diagnoses given the clinical context of West Africa. Urgent clinical correlation with patient symptoms, vital signs, and sputum/laboratory studies is essential to guide appropriate management. Immediate assessment for hypoxemia and respiratory support is warranted.

■ **DISCLAIMER:** This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.